

Raw Material Knowledge Engine

Data Dictionary and Schema Reference

MAERII Project Team

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Raw Material Knowledge Engine

Data Dictionary and Schema Reference

MAERII: Market Alignment Engine for Rural and Indigenous Innovation

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This document provides the schema-level reference for the MAERII Raw Material Knowledge Engine. It explains how the dataset is structured, where values come from, how evidence is stored, and how graph relationships are built.

1 Table of Contents

Contents

1	Table of Contents	i
2	Purpose of This Document	1
3	Dataset Inventory	1
4	Data Construction Overview	2
5	Canonical Dataset Schema	3
5.1	Canonical Record Shape	3
5.2	Base Field Dictionary	3
5.3	Base Field Coverage	5
6	Deep Enrichment Schema	6
6.1	Deep Enrichment Record Shape	6
6.2	Deep Enrichment Field Dictionary	6
6.3	Quantity Record Schema	7
6.4	Deep Enrichment Coverage	8
6.5	Deep Status Distribution	8
6.6	Availability Band Distribution	8
7	Evidence Schema	8
7.1	Evidence Record Shape	9
7.2	Evidence Field Dictionary	9
7.3	Evidence Attribute Distribution	9
8	Source Classification and Locations	10
8.1	Internal Source Locations.....	10
8.2	External Source Types.....	10
8.3	Evidence Source Distribution.....	11
9	LLM Extraction Behavior	11
9.1	Base LLM Extraction.....	11
9.2	Deep LLM Extraction.....	12
9.3	Where the LLM Explores.....	12
9.4	LLM Output Handling.....	12
10	Relationship Edge Schema	13
10.1	Edge Record Shape.....	13
10.2	Edge Field Dictionary.....	13
10.3	Edge Type Dictionary.....	13
10.4	Relationship Edge Distribution.....	14
10.5	Target Type Distribution.....	14
11	How Relationships Are Built	14
12	Confidence Scoring Reference	15

13 Validation and Integrity Rules	15
14 Example: Tamarind Record	16
14.1 Canonical Material Context.....	16
14.2 Deep Enrichment Summary.....	16
14.3 Evidence Example.....	17
14.4 Relationship Edge Example.....	17
15 Schema Usage by Future Modules	17
16 Conclusion	18

2 Purpose of This Document

This document is the formal data dictionary and schema reference for the Raw Material Knowledge Engine. It is intended to explain the structure of the dataset in enough detail that a reviewer, developer, mentor, or future MAERII module owner can understand exactly what each file contains, what each field means, where the data came from, and how the system converts unstructured source material into structured records.

The first technical report explains the overall system architecture and project outcomes. This second document focuses on the dataset itself. It describes the canonical material schema, the deep enrichment schema, the evidence schema, the relationship edge schema, the source classification model, and the LLM's role in extraction.

The most important principle in this data model is traceability. The system does not treat the LLM as the final authority. Instead, the LLM is used as an extraction assistant over retrieved source text. Extracted values are stored with confidence scores and are linked to source evidence wherever possible.

3 Dataset Inventory

The Raw Material Knowledge Engine uses one canonical item dataset and several supporting sidecar files. The canonical dataset stores the material records. The sidecar files store evidence, graph relationships, validation summaries, source traces, and run artifacts.

File Location	Approx. Size	Role
data/enriched_mfp_data.json	280 KB	Canonical enriched MFP item dataset
data/deep_enrichment_evidence.json	747 KB	Evidence store keyed by <code>evidence_id</code>
data/material_relationships.json	313 KB	Graph-ready relationship edge list
data/seed_mfp_list.json	41 KB	Cleaned seed records derived from the original MFP CSV
data/enrichment_log.json	53 KB	Validation report for the base enrichment run
data/enrichment_sources.json	20 KB	Source tracking from the base enrichment stage
data/deep_enrichment_runs/	Variable	Run manifests, previews, and validation outputs for deep enrichment
data/backups/	Variable	Pre-apply snapshots used for rollback

The current dataset statistics are summarized below.

Metric	Value
Seed MFP records	87
Canonical enriched records	87
Records with deep_enrichment block	86

Metric	Value
Evidence records	1,502
Relationship edges	1,259
Fully complete base records	66
Average base completeness	96 percent
Base records needing review	72

4 Data Construction Overview

The complete dataset is built through two related enrichment paths: base enrichment and deep enrichment.

Base enrichment creates the core material profile. It begins with the TRIFED MFP seed list and adds fields such as scientific name, description, season, shelf life, states, artisan types, current products, potential products, image URL, and confidence values.

Deep enrichment extends the existing record without rewriting it. It adds a nested deep_enrichment object to the canonical item record and writes full source provenance to sidecar files. The deep enrichment path targets availability, quantities, districts, clusters, source evidence, and graph relationships.

The data construction flow is shown below.

Stage	Source	System Action	Output
Seed preparation	MFP_List_87_Items_Split.csv	Parse CSV, normalize MSP, expand category codes, clean scientific names	seed_mfp_list.json
Base search	Serper web and image search	Generate field-specific search queries for each material	Search result snippets and URLs
Base fetching	HTML pages	Fetch and sanitize page text	Aggregated source text
Base LLM extraction	Aggregated source text	Extract base fields as strict JSON	Base enriched item fields
Base validation	Enriched item records	Check missing fields, confidence, and structural issues	enrichment_log.json
Deep search	Existing enriched records	Search for quantity, geography, and relationship evidence	Ranked source list
Deep fetching	HTML and PDF documents	Extract text from webpages and PDFs	Source dossier
Deep LLM extraction	Source dossier	Extract availability and geography only	deep_enrichment patch
Evidence modeling	Extracted values and source snippets	Create deterministic evidence IDs	deep_enrichment_evidence.json
Relationship modeling	Existing fields plus deep geography	Build graph-ready edges	material_relationships.json

5 Canonical Dataset Schema

The canonical dataset is stored at `data/enriched_mfp_data.json`. It is a JSON array. Each object represents one Minor Forest Produce material.

The record contains both original seed attributes and enriched attributes. Deep enrichment is stored as a nested object named `deep_enrichment`. Existing top-level fields remain the protected baseline.

5.1 Canonical Record Shape

```
{
  "mfp_id": 1,
  "name": "Tamarind (with seeds)",
  "scientific_name": "Tamarindus indica",
  "category": "Forest Produce",
  "msp": 36.0,
  "unit": "kg",
  "msp_notes": null,
  "applicability_raw": "All India",
  "states": ["Tamil Nadu", "Madhya Pradesh"],
  "description": "Concise material description...",
  "season": "March-April",
  "shelf_life": "9-12 months",
  "artisan_types": ["Fruit Collector"],
  "current_products": ["pulp", "powder"],
  "potential_products": ["seed derivatives"],
  "image_url": "https://example.com/image.jpg",
  "confidence": {
    "description": 0.9,
    "states": 0.8
  },
  "deep_enrichment": {
    "version": "v1",
    "status": "partial"
  }
}
```

5.2 Base Field Dictionary

Field	Type	Required	Description	Example
mfp_id	integer	Yes	Stable numeric identifier assigned during seed parsing.	1
name	string	Yes	Common name of the MFP item from the source list.	Tamarind (with seeds)

Field	Type	Required	Description	Example
scientific_name	string or null	No	Botanical or scientific name, cleaned during seed preparation or enriched by LLM.	Tamarindus indica
category	string	Yes	Expanded category derived from source category codes.	Forest Produce
msp	number or null	No	Minimum Support Price value parsed from the source list.	36.0
unit	string	Yes	Unit associated with MSP, generally kilograms.	kg
msp_notes	string or null	No	Notes for complex MSP values that cannot be represented as a single number.	3200 / 1500 (per Thousand)
Applicability	string	Yes	Original applicability text from the seed source before normalization.	All India
states	array of strings	No	Indian states where the material is known or inferred to be collected or available.	["Tamil Nadu", "Madhya Pradesh"]
description	string or null	No	Concise description of the material and its relevance.	Tamarind is a leguminous tree...
season	string or null	No	Harvesting or collection season, preferably expressed as months.	March-April
shelf_life	string or null	No	Storage duration in short duration format.	9-12 months

Field	Type	Required	Description	Example
artisan_types	array of strings	No	Types of collectors, processors, or artisans associated with the material.	["Fruit Collector", "Seed Collector"]
current_products	array of strings	No	Products already made from the material.	["pulp", "powder"]
potential_products	array of strings	No	Higher-value or future value-added product possibilities.	["seed derivatives"]
image_url	string or null	No	Representative image URL retrieved through image search.	https://...
confidence	object	No	Field-level confidence values for base enrichment.	{"description": 0.9}
deep_enrichment	object	No	Additive deep enrichment block for supply intelligence and graph support.	See Section 5

5.3 Base Field Coverage

Field	Coverage
scientific_name	85 / 87
description	87 / 87
season	77 / 87
shelf_life	72 / 87
states	85 / 87
artisan_types	87 / 87
current_products	87 / 87
potential_products	81 / 87
image_url	87 / 87

6 Deep Enrichment Schema

The `deep_enrichment` object is an additive extension inside each canonical material record. It is designed to store extra supply-side intelligence without changing the base record.

The deep enrichment block captures four major concepts:

- availability and quantity signals
- granular geography, especially districts and clusters
- relationship summaries for downstream modules
- evidence references and confidence values

6.1 Deep Enrichment Record Shape

```
{
  "version": "v1",
  "status": "partial",
  "availability": {
    "band": "medium",
    "quantity_records": [
      {
        "value": 250000,
        "unit": "tonnes",
        "year": null,
        "scope": "annual production",
        "metric_type": "production"
      }
    ]
  },
  "geography": {
    "districts": [],
    "clusters": []
  },
  "relationship_summary": {
    "related_regions": ["Tamil Nadu", "Madhya Pradesh"],
    "related_products": ["pulp", "powder"],
    "related_skills": ["Fruit Collector"],
    "related_material_groups": ["Forest Produce"]
  },
  "confidence": {
    "availability": 0.7,
    "geography": 0.0,
    "relationships": 1.0
  },
  "evidence_refs": ["evi_363051b04e7b5c30"],
  "last_updated_at": "2026-06-21T08:38:25Z"
}
```

6.2 Deep Enrichment Field Dictionary

Field	Type	Description
version	string	Schema version for deep enrichment. Current value is v1.
status	string	Processing status. Expected values include not_started, partial, complete, and failed.
availability	object	Availability intelligence, including a broad availability band and optional numeric records.
availability.band	string	Conservative qualitative availability classification. Values are unknown, low, medium, or high.
availability.quantity_records	array of objects	Numeric quantity, production, procurement, collection, or yield records extracted from sources.
geography	object	Granular geographic details beyond broad state-level data.
geography.districts	array of strings	Districts explicitly or reasonably supported by source text.
geography.clusters	array of strings	Named livelihood, processing, tribal, or Van Dhan clusters.
relationship_summary	object	Inline summary of relationships for quick access by later modules.
relationship_summary.related_regions	array of strings	States, districts, or clusters related to the material.
relationship_summary.related_products	array of strings	Current and potential products related to the material.
relationship_summary.related_skills	array of strings	Artisan or processing skill labels related to the material.
relationship_summary.related_material_groups	array of strings	Broad material group or category labels.
confidence	object	Confidence values for availability, geography, and relationship construction.
evidence_refs	array of strings	Evidence IDs that support the deep enrichment block or relationship summary.
last_updated_at	string or null	UTC timestamp for the last additive enrichment update.

6.3 Quantity Record Schema

Quantity records are intentionally conservative. They are stored only when a numeric statement is found in the retrieved evidence.

Field	Type	Description	Example
value	number	Numeric quantity value.	250000
unit	string	Normalized unit. Common values include tonnes, kg, and quintals.	tonnes
year	integer or null	Year associated with the value, if available.	2005

Field	Type	Description	Example
scope	string or null	Short phrase describing the scope of the value.	annual production
metric_type	string or null	Type of quantity signal. Expected values include collection, procurement, production, and yield.	production

6.4 Deep Enrichment Coverage

Deep Field	Coverage
Records with deep_enrichment	86 / 87
Records with quantity records	41 / 87
Records with districts	38 / 87
Records with clusters	24 / 87
Records with relationship summary	86 / 87

6.5 Deep Status Distribution

Status	Count
partial	65
not_started	21

6.6 Availability Band Distribution

Band	Count
medium	14
low	2
unknown	70

7 Evidence Schema

Evidence records are stored in `data/deep_enrichment_evidence.json`. This file is a JSON object keyed by `evidence_id`. Each evidence record explains where a particular value came from and how strongly it is supported.

The evidence store is deliberately separate from the canonical item file. This keeps `enriched_mfp_data.json` readable while preserving detailed provenance in a dedicated reference file.

7.1 Evidence Record Shape

```
{
  "evidence_id": "evi_363051b04e7b5c30",
  "mfp_id": 1,
  "attribute_group": "availability",
  "attribute_name": "band",
  "value": "low",
  "source_url": "https://gef7folur.da.gov.in/wwwroot/Doc/ProDoc.pdf",
  "source_type": "official",
  "snippet": "Compound annual growth rates of area, production and yield...",
  "retrieved_at": "2026-06-21T08:38:25Z",
  "confidence": 0.3
}
```

7.2 Evidence Field Dictionary

Field	Type	Description
evidence_id	string	Deterministic evidence identifier generated from material ID, attribute, value, source URL, and snippet.
mfp_id	integer	Material ID to which the evidence belongs.
attribute_group	string	Broad group supported by the evidence. Examples: availability, geography, relationship.
attribute_name	string	Specific attribute supported by the evidence. Examples: band, quantity_records, districts, state, skill.
value	string, number, object, or array	The extracted or derived value supported by the evidence.
source_url	string	URL or synthetic dataset reference from which the evidence was derived.
source_type	string	Source category assigned by the system.
snippet	string	Short text excerpt or explanation supporting the value.
retrieved_at	string	UTC timestamp indicating when the evidence record was created.
confidence	number	Confidence score between 0.0 and 1.0.

7.3 Evidence Attribute Distribution

Attribute Group	Count
relationship	1,206
availability	194
geography	102

Attribute Name	Count
state	390
current_product	317

Attribute Name	Count
potential_product	236
skill	177
band	113
material_group	86
quantity_records	81
districts	71
clusters	31

8 Source Classification and Locations

The dataset is built from multiple kinds of locations. Some are direct data files inside the project. Others are external web sources discovered through search and document retrieval.

8.1 Internal Source Locations

Location	Purpose
MFP_List_87_Items_Split.csv	Original seed list used to create material records.
data/seed_mfp_list.json	Cleaned seed representation used by the enrichment pipeline.
data/enriched_mfp_data.json	Canonical enriched dataset.
data/enrichment_sources.json	Base enrichment source tracking.
data/deep_enrichment_evidence.json	Evidence records for deep enrichment and relationships.
data/material_relationships.json	Graph-ready relationship edge list.
data/deep_enrichment_runs/	Run manifests, previews, and validation summaries.
data/backups/	Snapshots used for rollback and recovery.

8.2 External Source Types

The deep enrichment system classifies source URLs into trust categories. Source classification is implemented in `mfp_scraper/deep_models.py`.

Source Type	Meaning	Typical Domain Hints
official	Government or official public-sector source.	gov.in, nic.in
institutional	Biodiversity, forestry, agricultural, or institutional source.	gbif.org, powo.science.kew.org, indiabiodiversity.org, icfre.gov.in, icar.gov.in, nmpb.nic.in

Source Type	Meaning	Typical Domain Hints
research	Research paper, journal, repository, or academic source.	ncbi.nlm.nih.gov, researchgate.net, sciencedirect.com, springer.com, mdpi.com
supporting	General web source used as secondary support.	Other web domains
existing_dataset	Synthetic source reference derived from existing canonical fields.	dataset://enriched_mfp_data.json

8.3 Evidence Source Distribution

Source Type	Evidence Records
existing_dataset	1,206
official	175
research	74
supporting	47

The large number of existing_dataset records is expected. Many graph relationships, such as material-to-state or material-to-product edges, are derived from already validated canonical fields rather than from newly retrieved web documents.

9 LLM Extraction Behavior

The LLM is used as a structured extraction assistant. It is not used to freely invent facts or decide the final dataset independently. The system retrieves source text first, creates a bounded source dossier, and then asks the LLM to extract specific fields in a strict JSON shape.

9.1 Base LLM Extraction

Base enrichment uses the prompt defined in mfp_scraper/llm_extractor.py. The LLM receives the material name, category, existing scientific name, MSP, known states, and aggregated source text.

The base prompt asks the LLM to extract:

- scientific name
- description
- season
- shelf life
- states
- artisan types
- current products
- potential products
- confidence scores

The prompt includes constraints such as using specific Indian states instead of “All India,” keeping shelf life as a short duration, and returning only valid JSON.

9.2 Deep LLM Extraction

Deep enrichment uses the prompt defined in `mfp_scraper/deep_extractor.py`. This prompt is narrower than the base prompt. It extracts only availability and geography fields from the source dossier.

The deep prompt asks the LLM to extract:

- `availability.band`
- `availability.quantity_records`
- `geography.districts`
- `geography.clusters`
- supporting evidence entries
- confidence values for availability and geography

The LLM is explicitly instructed not to infer exact numeric quantities when the source does not state them. It is also instructed not to place whole states in the district list. This prevents broad geography from being incorrectly treated as granular location data.

9.3 Where the LLM Explores

The LLM itself does not browse the web. Exploration is performed before the LLM call by the pipeline.

The system explores the web through targeted Serper search queries. For each material, the deep pipeline generates three query families:

Query Family	Purpose	Example Pattern
quantity	Find production, collection, yield, or quantity claims.	"Tamarind (with seeds)" India annual production OR collection OR yield tonnes
geography	Find producing districts or clusters.	"Tamarind (with seeds)" major producing districts India
relationships	Find processing, value addition, and product links.	"Tamarind (with seeds)" tribal value addition processing products

After search results are returned, the pipeline ranks sources by trust category, fetches the top documents, extracts text from HTML or PDF files, and passes the resulting source dossier to the LLM.

9.4 LLM Output Handling

The LLM output is parsed as JSON. If the model returns markdown fences or extra formatting, the parser attempts to recover the JSON object. After parsing, the output is normalized:

- quantity records are deduplicated using value, unit, year, scope, and metric type
- districts and clusters are deduplicated case-insensitively

- invalid availability bands are reset to unknown
- unsupported source URLs are discarded during normalization
- missing or unreliable values are left empty rather than guessed

10 Relationship Edge Schema

Relationship edges are stored in data/material_relationships.json. This file is a JSON array. Each object represents one directed graph edge from a material to another entity.

10.1 Edge Record Shape

```
{
  "edge_id": "edge_d7e284ed9506b1f9",
  "source_mfp_id": 1,
  "edge_type": "available_in",
  "target_type": "state",
  "target_value": "Tamil Nadu",
  "evidence_id": "evi_3be473ddd45f3573",
  "confidence": 1.0
}
```

10.2 Edge Field Dictionary

Field	Type	Description
edge_id	string	Deterministic edge identifier generated from source material, edge type, target type, target value, and evidence ID.
source_mfp_id	integer	MFP material ID from which the relationship originates.
edge_type	string	Type of relationship being represented.
target_type	string	Type of target entity. Examples include state, district, skill, current_product, and potential_product.
target_value	string	Human-readable target entity value.
evidence_id	string	Evidence record supporting the relationship.
confidence	number	Confidence score between 0.0 and 1.0.

10.3 Edge Type Dictionary

Edge Type	Target Types	Meaning
available_in	state, district	Indicates that a material is available, collected, produced, or associated with a region.
linked_to	cluster	Links a material to a cluster or institutional grouping.
processed_by	skill	Links a material to artisan, collector, or processor skill types.
used_for_product	current_product	Links a material to an existing product.
could_enable_product	potential_product	Links a material to a possible value-added product.
belongs_to_group	material_group	Links a material to its category or broader material class.

10.4 Relationship Edge Distribution

Edge Type	Count
available_in	426
processed_by	177
used_for_product	317
could_enable_product	236
belongs_to_group	86
linked_to	17

10.5 Target Type Distribution

Target Type	Count
state	390
district	36
skill	177
current_product	317
potential_product	236
material_group	86
cluster	17

11 How Relationships Are Built

The relationship graph is built primarily in `mfp_scraper/deep_pipeline.py`. The system uses both existing canonical fields and new deep enrichment fields.

Relationships are created deterministically. This means the graph is not left to the LLM to invent. The LLM helps identify availability and geography evidence, but most relationship edges are constructed by explicit rules.

The relationship construction process is as follows.

1. For every state in `states`, create an `available_in` edge to a state target.
2. For every district found in deep geography evidence, create an `available_in` edge to a district target.
3. For every cluster found in deep geography evidence, create a `linked_to` edge to a cluster target.
4. For every artisan type in `artisan_types`, create a `processed_by` edge to a skill target.
5. For every current product in `current_products`, create a `used_for_product` edge to a `current_product` target.
6. For every potential product in `potential_products`, create a `could_enable_product` edge to a `potential_product` target.
7. For the material category, create a `belongs_to_group` edge to a `material_group` target.

Each generated edge receives an `edge_id` and an `evidence_id`. When the edge comes from an existing canonical field, the evidence source is synthetic and uses the format `dataset://enriched_mfp_data.json#mfp_id=<id>`. When the edge comes from newly retrieved geography evidence, the evidence source points to the external document URL.

12 Confidence Scoring Reference

Confidence scores are numeric values between 0.0 and 1.0. They should be interpreted as system confidence in the extracted or derived value, not as statistical probability.

Range	Interpretation
0.0	No evidence or not populated.
0.1 - 0.4	Weak or limited support. Use for review only.
0.5 - 0.7	Moderate support from source text or extracted evidence.
0.8 - 0.9	Strong support from explicit source material.
1.0	Directly derived from existing canonical dataset fields or otherwise treated as deterministic.

In the relationship graph, many edges have confidence 1.0 because they are deterministic transformations of existing canonical fields. This does not mean the real-world fact can never be wrong. It means the relationship edge accurately represents what is already present in the canonical dataset.

13 Validation and Integrity Rules

Validation is performed at two levels: base validation and deep validation.

Base validation checks the canonical enrichment fields. It reports completeness, missing fields, low-confidence fields, and potential issues such as excessive state lists or overlap between current and potential products.

Deep validation checks the relationship between the canonical dataset and the sidecar files. It verifies that evidence references exist, relationship edges point to valid material IDs, and edge evidence IDs are present in the evidence store.

The main integrity rules are:

- mfp_id must remain stable across all files
- every deep_enrichment.evidence_refs entry should exist in deep_enrichment_evidence.json
- every relationship edge should point to an existing material ID
- every relationship edge should have a valid evidence_id
- existing top-level canonical fields should not be changed by deep enrichment
- deep enrichment should remain additive and reversible

14 Example: Tamarind Record

The following example demonstrates how one material record is represented across the canonical dataset, evidence store, and relationship graph.

14.1 Canonical Material Context

Field	Value
mfp_id	1
name	Tamarind (with seeds)
scientific_name	Tamarindus indica
category	Forest Produce
states	Tamil Nadu, Madhya Pradesh, Andhra Pradesh, Maharashtra, Karnataka
season	March-April
shelf_life	9-12 months

14.2 Deep Enrichment Summary

Field	Value
status	partial
availability.band	medium
availability.quantity_records	3 records
geography.districts	0 records
geography.clusters	0 records
confidence.availability	0.7
confidence.geography	0.0
confidence.relationships	1.0

One quantity record for Tamarind is shown below.

```
{  
  "value": 250000,  
  "unit": "tonnes",  
  "year": null,  
  "scope": "annual production",  
}
```

```

    "metric_type": "production"
}

```

14.3 Evidence Example

```

{
  "evidence_id": "evi_adc55b0a0de69420",
  "mfp_id": 1,
  "attribute_group": "availability",
  "attribute_name": "quantity_records",
  "value": "250000 tonnes",
  "source_url": "https://journalcra.com/sites/default/files/issue-pdf/49267.pdf",
  "source_type": "supporting",
  "snippet": "Extensive tamarind orchards in India produce 250,000 tonnes...",
  "retrieved_at": "2026-06-21T08:38:25Z",
  "confidence": 0.8
}

```

14.4 Relationship Edge Example

```

{
  "edge_id": "edge_d7e284ed9506b1f9",
  "source_mfp_id": 1,
  "edge_type": "available_in",
  "target_type": "state",
  "target_value": "Tamil Nadu",
  "evidence_id": "evi_3be473ddd45f3573",
  "confidence": 1.0
}

```

15 Schema Usage by Future Modules

The Data Dictionary is not only a reference for the current module. It also defines how later MAERII modules should use the knowledge base.

Future Module	Relevant Fields
Market Demand Intelligence	current_products, potential_products, material_relationships.json
Product Recommendation AI	states, artisan_types, availability, geography, product edges
Product Design Enhancement AI	current_products, potential_products, artisan_types, material group
Quality Grading	category, description, shelf_life, source evidence
Pricing Engine	msp, unit, availability.band, quantity records, product relationships

The canonical item file should be used when a module needs a material profile. The relationship sidecar should be used when a module needs connected reasoning, such as finding all materials available in a state or all products linked to a material group. The evidence file should be used when a module or human reviewer needs to verify where a value came from.

16 Conclusion

The Raw Material Knowledge Engine dataset is structured around a clear separation of concerns. The canonical dataset stores material records, the evidence store preserves provenance, and the relationship file represents the knowledge graph layer. This structure allows the project to remain readable, auditable, and extensible.

The LLM's role is controlled and narrow. It extracts structured values from retrieved source text, while deterministic code handles normalization, deduplication, evidence linking, and relationship construction. This design gives MAERII a dataset that is not merely enriched, but explainable.